

Synthetic Geometry

Lines and Transformations

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"I saw an angle in the marble and carved until I set him free."

— Michelangelo

1 General Advice

1. Draw multiple diagrams, with different angle ratios.
2. Draw what you are trying to prove: the circles around cyclic quads, lines when proving concurrency etc.
3. Think of different reformulations of a condition, or similar configurations and write them down.
4. Dissect the problem into smaller lemmas to simplify it. Adding points and using affine transformations should become your best friends.

2 Group Solving

- **USAMO25/4** Let H be the orthocenter of acute triangle ABC , let F be the foot of the altitude from C to AB , and let P be the reflection of H across BC . Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y . Prove that C is the midpoint of XY .

A hint will be given every 15 minutes.

3 Problems

- **USAMO23/1** In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.
- **IndiaCamp12/1** Let ABC be an isosceles triangle with $AB = AC$. Let D be a point on the segment BC such that $BD = 2DC$. Let P be a point on the segment AD such that $\angle BAC = \angle BPD$. Prove that $\angle BAC = 2\angle DPC$.
- **IMO20/1** Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB .

- **IMO17/4** Let R and S be different points on a circle Ω such that RS is not a diameter. Let ℓ be the tangent line to Ω at R . Point T is such that S is the midpoint of the line segment RT . Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects ℓ at two distinct points. Let A be the common point of Γ and ℓ that is closer to R . Line AJ meets Ω again at K . Prove that the line KT is tangent to Γ .
- **ISL17/G4** In triangle ABC , let ω be the excircle opposite to A . Let D, E and F be the points where ω is tangent to BC, CA , and AB , respectively. The circle AEF intersects line BC at P and Q . Let M be the midpoint of AD . Prove that the circle MPQ is tangent to ω .
- **IMO18/1** Let Γ be the circumcircle of acute triangle ABC . Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.
- **TSTST17/1** Let ABC be a triangle with circumcircle Γ , circumcenter O , and orthocenter H . Assume that $AB \neq AC$ and that $\angle A \neq 90^\circ$. Let M and N be the midpoints of sides AB and AC , respectively, and let E and F be the feet of the altitudes from B and C in $\triangle ABC$, respectively. Let P be the intersection of line MN with the tangent line to Γ at A . Let Q be the intersection point, other than A , of Γ with the circumcircle of $\triangle AEF$. Let R be the intersection of lines AQ and EF . Prove that $PR \perp OH$.

4 Vague Hints

1. **ISL17/G4** – Translate the tangency condition, a certain operation will scream at you.
2. **IMO18/1** – Cyclic trapezoids.

Projective Geometry

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April 9, 2026

1 Group Solving

- **USATST13 Jan/2** Let ABC be an acute triangle. Circle ω_1 , with diameter AC , intersects side BC at F (other than C). Circle ω_2 , with diameter BC , intersects side AC at E (other than C). Ray AF intersects ω_2 at K and M with $AK < AM$. Ray BE intersects ω_1 at L and N with $BL < BN$. Prove that lines AB , ML , NK are concurrent.
- **Alexander Remorov** Point M lies on diagonal BD of parallelogram $ABCD$. Line AM intersects side CD and line BC at points K and N , respectively. Let C_1 be the circle with center M and radius MA and C_2 be the circumcircle of triangle $\triangle KCN$. C_1 , C_2 intersect at P and Q . Prove that MP , MQ are tangent to C_2 .

A hint will be given every 15 minutes.

2 Problems

- **USAJMO11** Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that (i) lines PB and PD are tangent to ω , (ii) P, A, C are collinear, and (iii) $DE \parallel AC$. Prove that BE bisects AC .
- **USAJMO15** Let $ABCD$ be a cyclic quadrilateral. Prove that there exists a point X on segment BD such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$ if and only if there exists a point Y on segment AC such that $\angle CBD = \angle YBA$ and $\angle CDB = \angle YDA$.
- **IMO85** A circle with centre O passes through vertices A, C of $\triangle ABC$ and intersects its sides BA, BC at distinct points K, N , respectively. The circumcircles of $\triangle ABC$ and $\triangle KBN$ intersect at point B and another point M . Prove that $\angle OMB = 90$.
- **Iran15** In quadrilateral $ABCD$, AC is the bisector of $\angle A$ and $\angle ADC = \angle ACB$. Let X and Y be the feet of perpendiculars from A to BC and CD , respectively. Prove that the orthocenter of triangle $\triangle AXY$ is on BD .
- **Vietnam03** Circles C_1 and C_2 are externally tangent at M , and radius of C_2 is greater than radius of C_1 . A is a point on C_2 which does not lie on the line joining the centers of the circles. Let B and C be points on C_1 such that AB and AC are tangent to C_1 . Lines BM and CM intersect C_2 again at E and F , respectively. Let D be the intersection of the tangent to C_2 at A and line EF . Show that the locus of D as A varies is a straight line.
- **APMO08** Let Ω be the circumcircle of a triangle ABC . A circle passing through points A and C meets the sides BC and BA at D and E , respectively. The lines AD and CE meet Ω again at G and H , respectively. The tangent lines of Ω at A and C meet the line DE at L and M , respectively. Prove that the lines LH and MG meet at Ω .
- **APMO13** Let $ABCD$ be a quadrilateral inscribed in a circle ω , and let P be a point on the extension of AC such that PB and PD are tangent to ω . The tangent at C intersects PD at Q and the line AD at R . Let E be the second point of intersection between AQ and ω . Prove that B, E, R are collinear.

- **IMO10** Given a triangle ABC , with I as its incenter and Ω as its circumcircle, AI intersects Ω again at D . Let E be a point on the arc BDC , and F a point on the segment BC , such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. If G is the midpoint of IF , prove that the meeting point of the lines EI and DG lies on Ω .

Problem Solving Session

April 2026

1 Problems

1. **PAMO17/6** Let ABC be a triangle with H its orthocenter. The circle with diameter $[AC]$ cuts the circumcircle of triangle ABH at K . Prove that the point of intersection of the lines CK and BH is the midpoint of the segment $[BH]$.
2. **PolishMO18** An acute triangle $\triangle ABC$ in which $AB < AC$ is given. Points E and F are feet of its heights from B and C , respectively. The line tangent in point A to the circle escribed on $\triangle ABC$ crosses BC at P . The line parallel to BC that goes through point A crosses EF at Q . Prove PQ is perpendicular to the median from A of triangle ABC .
3. **APMO2020/1** Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC . The tangent to Γ at A intersects the parallel line to BA through D at point E . The segment $[CE]$ intersects Γ again at F . Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

2 Hints

1. There's an angle chase solution, but the one I want you to find is one where you reflect one of the triangle's points to then try and find a parallelogram that includes $[BH]$.
2. Create a circle with radius zero.
3. Use Radical Center.

Problem Solving Session

April 17, 2026

1 Problems

1. **ISL23/G2** Let ABC be a triangle with $AC > BC$, let ω be the circum-circle of $\triangle ABC$, and let r be its radius. Point P is chosen on $\overline{AC}$ such that $BC = CP$, and point S is the foot of the perpendicular from P to $\overline{AB}$. Ray BP meets ω again at D . Point Q is chosen on line SP such that $PQ = r$ and S, P, Q lie on a line in that order. Finally, let E be a point satisfying $\overline{AE} \perp \overline{CQ}$ and $\overline{BE} \perp \overline{DQ}$. Prove that E lies on ω .
2. **ISL23/G3** Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A . Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$. Prove that lines AD, PM , and BC are concurrent.

AOA World Tour

Asia

April 2026

1 Problems

1. **TaiwanMO26** Let $\triangle ABC$ be inscribed in a circle ω . Denote M as the midpoint of BC . Let P be the midpoint of the chord of ω that passes through M and is perpendicular to AC . Let Q be the midpoint of the chord of ω that passes through M and is perpendicular to AB . Prove that PQ is perpendicular to AO , where O is the center of ω .
2. **VietnamMO26** Let ABC be an acute triangle ($AB < AC$) with circumcenter O . Let E, F be the reflection of A across B, C respectively. Let M be the midpoint of BC . Points I, P are chosen on line AM such that triangle AIE, APF are isosceles triangles with $IA = IE, PA = PF$. Let T be the intersection of IE and PF .
 - a) Prove that T lies on the circumcircle of triangle ABC .
 - b) Let D be the intersection of lines BI and CP . Let J be the intersection of lines TD and IP . Let X be the intersection of line OT and BC . Lines AT, AX intersect EF at L, U respectively. Prove that the circumcircle of triangles TAJ, TIP, TEF, TLU all have a common point other than T .
3. **ChinaTST26/1[3]** In an acute triangle ABC with $AB \neq AC$, let D be a point on the circumcircle such that $AD \perp BC$. Points P, Q lie inside $\triangle ABC$ satisfying

$$\angle APB = \angle APC = 180^\circ - \angle ACB,$$

$$\angle AQB = \angle AQC = 180^\circ - \angle ABC.$$

Prove that A, P, Q, D are concyclic.

4. **IndiaMO26** Let ABC be an acute-angled scalene triangle with circumcircle Γ . Let M be the midpoint of BC and N be the midpoint of the minor arc BC of Γ . Points P and Q lie on segments AB and AC respectively such that $BP = BN$ and $CQ = CN$. Point $K \neq N$ lies on line AN with $MK = MN$. Prove that $\angle PKQ = 90^\circ$.

Pretend this Title Connects the Two Topics Here

Raihane

April 2026

1 Humpty and Dumpty

1.1 Humpty point

Original Definition In triangle ABC the A -Humpty point H_A is defined to be a point inside triangle such that $\angle H_A BC = \angle H_A AB$ and $\angle H_A CB = \angle H_A AC$.

Alternative Definition (better for contests) The A -Humpty Point, H_A , is the intersection of (HA) and (HBC) Where H is the Orthocenter of $\triangle ABC$.

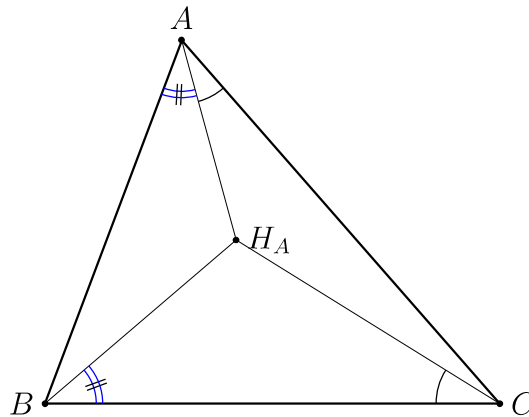


FIGURE 1. A -Humpty Point

Facts about H_A :

1. Lies on A median of ABC .
2. Lies on A Apollonius circle of ABC that is, $\frac{AB}{AC} = \frac{H_AB}{H_AC}$.
3. B, H_A, H, C are concyclic, where H is the orthocentre of ABC .
4. $HH_A \perp AH_A$.

1.2 Dumpty point

Definition: In triangle ABC , the A Dumpty point D_A is defined to be a point inside triangle such that $\angle D_A B A = \angle D_A A C$ and $\angle D_A A B = \angle D_A C A$.

Alternate Definition (Better for Contests): The A -Dumpty Point is the inter-section of (AO) and (BOC) . Where O is the circumcenter of $\triangle ABC$

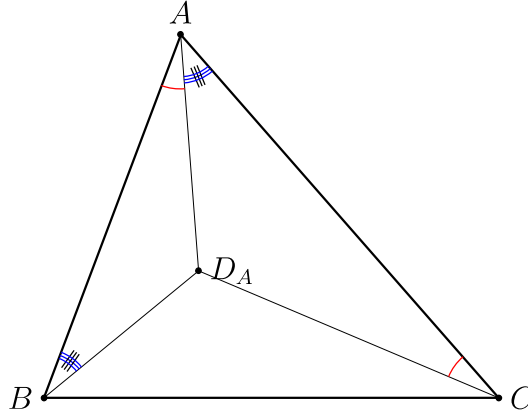


FIGURE 2. A -Dumpty point

Facts about D_A :

1. Lies on A symmedian.
2. It is the centre of spiral similarity sending $\triangle AD_A C$ to $\triangle CD_A B$, i.e., sending AC to BA .
3. B, D_A, O, C are concyclic where O is circumcentre of $\triangle ABC$.
4. $OD_A \perp AD_A$.
5. H_A and D_A are isogonal conjugates.

1.3 Problems for this section

1. **Vietnam2026/4** Let ABC be an acute triangle ($AB < AC$) with circumcenter O . Let E, F be the reflection of A across B, C respectively. Let M be the midpoint of BC . Points I, P are chosen on line AM such that triangle AIE, APF are isosceles triangles with $IA = IE, PA = PF$. Let T be the intersection of IE and PF .
 - a) Prove that T lies on the circumcircle of triangle ABC .
 - b) Let D be the intersection of lines BI and CP . Let J be the intersection of lines TD and IP . Let X be the intersection of line OT and BC . Lines AT, AX intersect EF at L, U respectively. Prove that the circumcircle of triangles TAJ, TIP, TEF, TLU all have a common point other than T .

2. **India EGMOTST25/5** Let acute scalene $\triangle ABC$ have circumcircle ω . Let M be the midpoint of BC , define X as the other intersection of AM with ω . Let E, F be the feet of altitudes from B, C to AC, AB respectively. Let Q be the second intersection of the circumcircle of $\triangle AEF$ and ω . Let $Y \neq X$ be a point such that $MX = MY$ and QMY is cyclic. Finally, let S be a point on BC such that $\angle BAS = \angle MAC$.

Prove that the quadrilaterals $BFYS$ and $CEYS$ are cyclic.

2 Similar Quadrilaterals

(**Triangles form Similar Quads**) Let $\triangle ABC \sim \triangle XYZ$ and $\triangle ABD \sim \triangle XYW$ (both similarities in the same orientation). Prove that quadrilateral $ABCD$ is similar to quadrilateral $XYZW$ ($ABCD \sim XYZW$) (again both in the same orientation).

(**SSAAA Similarity**) Two quadrilaterals $ABCD$ and $A'B'C'D'$ satisfy $\angle A = \angle A'$, $\angle B = \angle B'$, $\angle C = \angle C'$, and $\frac{AB}{A'B'} = \frac{BC}{B'C'}$. Show that $ABCD$ and $A'B'C'D'$ are similar.

2.1 Problems for this section

1. **CentroAmerican23** Let ABC be an acute-angled triangle with $AB < AC$ and Γ the circumference that passes through A, B and C . Let D be the point diametrically opposite A on Γ and ℓ the tangent through D to Γ . Let P, Q and R be the intersection points of BC with ℓ , of AP with Γ such that $Q \neq A$ and of QD with the A -altitude of the triangle ABC , respectively. Define S to be the intersection of AB with ℓ and T to be the intersection of AC with ℓ . Show that S and T lie on the circumference that passes through A, Q and R .
2. **IMO04/5** In a convex quadrilateral $ABCD$, the diagonal BD bisects neither the angle ABC nor the angle CDA . The point P lies inside $ABCD$ and satisfies

$$\angle PBC = \angle DBA \quad \text{and} \quad \angle PDC = \angle BDA.$$

Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.

3. **IndiaMO14** In an acute-angled triangle ABC , a point D lies on the segment BC . Let O_1, O_2 denote the circumcenters of triangles ABD and ACD respectively. Prove that the line joining the circumcenter of triangle ABC and the orthocenter of triangle O_1O_2D is parallel to BC .